

Antara Ganapathy

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Summary

Dedicated first-year BSc Computer Science student with a passion for tech-entrepreneurship and fintech. Having gained different skills and experience in the field, I aim to leverage my tech knowledge to drive social impact.

Education

The University of Manchester, BSc in Computer Science Sept 2024 – June 2027

- First-year courses: Mathematical Techniques for Computer Science, Fundamentals of Computer Engineering, Fundamentals of Computer Architecture, and Introduction to Programming I & II, Fundamentals of Computation, Data Science, Operating Systems

Indus International School, Bangalore, IB Diploma Programme Sept 2022 – May 2024

- Subjects taken: Computer Science HL, Math Analysis and Approaches HL, Physics HL, Economics SL, English Lang & Lit SL, French AB Initio SL

Experience

Spring into Software Engineering April 2025
JPMorganChase, Glasgow

- Attended panel discussions, networking sessions and workshops over the week to learn about Tech at JPMC
- Gained an in-depth understanding of the lines of businesses at JPMC and the role of tech in each
- Participated in the Code for Good Hackathon by developing PathMatch to improve the mentor registration and matching algorithm for MCR Pathways
- Developed PathMatch with React, Django, Tailwind & Gemini API

Robotics Society Jan 2025 - April 2025
University of Manchester

- Participated in the Buggy project - received an award for Best Effort with teammates
- Used Arduino to create a autonomous Buggy capable of navigating a maze with an ultrasonic sensor.
- Participated in Hack-a-Bot - worked on developing a robot to safely transport toxic waste.
- Arduino and HC-05 Bluetooth Module used to control the robot through Bluetooth Terminal

Budgetary Jan 2022 - Feb 2024

- Developed a mobile app with Flutter / Dart that aims to help university students manage their money effectively.
- Released a beta version which included features such as expense logs, currency converter (live exchange rates via an API), and to-do lists.
- Developed web-app with Python Flask to help instill financial discipline in children; briefly deployed with Heroku
- 1 of 3 highschoolers selected across India for pre-incubation program by Runway

Projects

PathMatch Github

- Created PathMatch with my team as part of the Code for Good Hackathon held by JPMorganChase
- A streamlined service that aims to improve the mentor registration and matching algorithm for MCR Pathways.
- Developed with ReactJS, Tailwind, Django and Gemini API

Hack The Code Github

- Competition hosted by Reply
- Used Python to optimize the consumption of green energy resources

- Was in the top 20% scoring teams

RAG Chatbot

Github

- Developed a chatbot with Retrieval-Augmented Generation to allow users to chat with a PDF
- Used Pinecone vector database and LangChain to develop the chatbot and Python Streamlit to develop the web-app

Sentiment Analysis

Github

- Trained a model with Natural Language Processing to analyze the sentiment news headlines
- Used web-scraping, with Python, to generate a test set of CNN news headlines

Glaucoma Detection

Github

- Trained a model to detect glaucoma on the 'Glaucoma Detection' Kaggle dataset
- Compared the performance of different pre-trained models: VGG, ResNet, Inception, and Xception
- 1/15 (of 200) students selected to present at the final round of the National High School Data Science Competition (NDSC), conducted in partnership with the Harvard Undergraduate Open Data Project

Diabetic Retinopathy Detection

Github

- Trained a model to detect diabetic retinopathy and the risk of macular edema on the Indian Diabetic Retinopathy Image Dataset (IDRiD).
- Investigated the impact of transfer learning with different pre-trained models such as InceptionNet and VGG16 on the model trained using a CNN.

Fake News Detection

Github

- Trained a model using NLP to detect fake news using the 'Fake and real news dataset'.
- Implemented techniques such as the TF-IDF vectorizer and the multinomial naive Bayes classifier

Publications

Defining Evaluative Metrics for Medical Imaging Datasets

May 2023

Young, Scientist Journal Vanderbilt University

[Journal Link](#)

Technologies

Python (Flask + AI/ML), Flutter/Dart, HTML, CSS (Tailwind), JavaScript (React), SQL